

Functional enrichment and regulatory factor associations of H3K4me1 vs H3K4me3 peaks

Pathway-level enrichment of genes proximal to CUT&Tag peaks in C1 showed a pronounced bias toward H3K4me3-associated programs (Figure 1). Genes near H3K4me3 peaks were most strongly enriched for P53_PATHWAY and DNA_REPAIR, with additional robust signals in MYC_TARGETS_V1, G2M_CHECKPOINT, and E2F_TARGETS. More moderate associations were detected for MTORC1_SIGNALING, OXIDATIVE_PHOSPHORYLATION, UNFOLDED_PROTEIN_RESPONSE, and APICAL_JUNCTION, whereas many other pathways (e.g., INTERFERON_GAMMA_RESPONSE, KRAS_SIGNALING_UP, WNT_BETA_CATENIN_SIGNALING) showed minimal or no enrichment. By contrast, genes proximal to H3K4me1 peaks displayed uniformly weaker enrichment across these gene sets. Collectively, these results indicate that, in C1, promoter-proximal H3K4me3 preferentially captures p53/DNA damage-response and cell-cycle/proliferation circuits, with coordinated engagement of MYC- and E2F-linked programs.

In parallel, transcription factor (TF) motif enrichment across the same peak sets indicated broad TF occupancy with limited mark-specific polarization (Figure 2). Most TF motifs were enriched in both H3K4me1- and H3K4me3-defined regions, with only scattered asymmetries and few isolated depletions. The lack of coherent TF groups segregating by mark suggests that the major regulatory factors in C1 occupy both enhancer-like (H3K4me1) and promoter-like (H3K4me3) chromatin, and that TF motif over-representation is largely shared rather than dictated by mark identity.

Together, these findings support a model in which H3K4me3-marked promoters in C1 preferentially encode p53/DNA-repair and cell-cycle/MYC/E2F transcriptional programs, while underlying TF occupancy is broadly shared across promoter- and enhancer-associated chromatin without strong mark-specific partitioning. This configuration aligns with

established roles of H3K4me3 in promoter-associated activation of proliferative and genome-maintenance pathways (Camacho & Fallahi-Sichani, 2026).

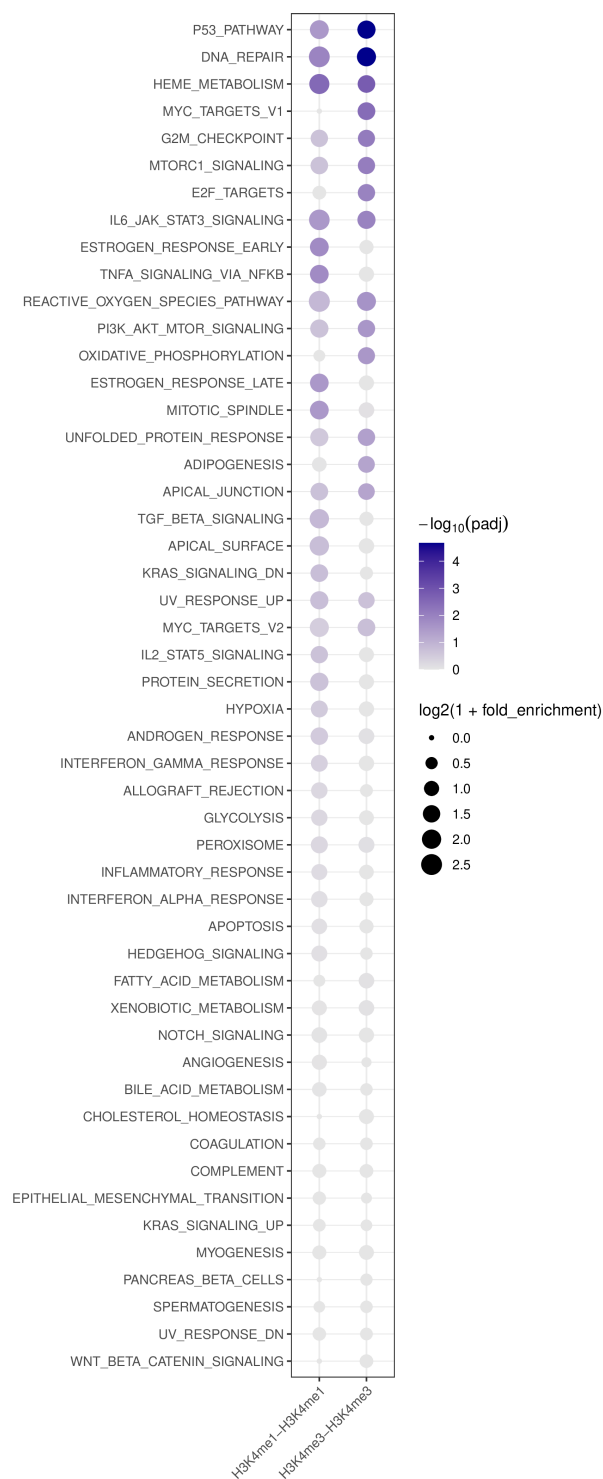


Figure 1. Pathway enrichment for genes near H3K4me1 and H3K4me3 peaks. Bubble plot of hallmark gene set enrichment for genes proximal to CUT&Tag peaks in sample C1. Signals concentrate for H3K4me3, with strong enrichment in P53 pathway, DNA repair, MYC targets, G2M checkpoint, and E2F targets, while H3K4me1 shows little to no enrichment across pathways.

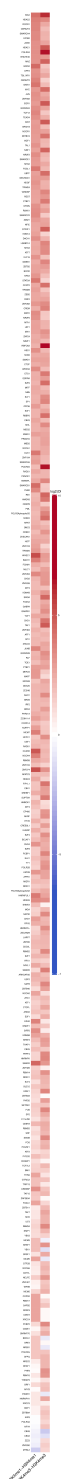


Figure 2. TFBS enrichment across H3K4me1 and H3K4me3 peaks. *Heatmap summarizing transcription factor binding or motif overlap enrichments for peaks derived from H3K4me1 and H3K4me3 in hg38. Enrichment is largely positive for both peak sets, with scattered mark-specific differences and occasional depletions.*

Regulatory context of peaks: cCRE classes and genic feature distributions

We annotated peaks in the hg38 coordinate system by intersecting them with the ENCODE cCRE registry and by assigning genic features to quantify regulatory and genomic context. The cCRE composition revealed clear mark-specific biases (Figure 3). H3K4me1 peaks were predominantly assigned to enhancer-like elements, with distal enhancer-like signatures (dELS) and proximal enhancer-like signatures (pELS) comprising the majority and only a small promoter-like signature (PLS) component. In contrast, H3K4me3 peaks were dominated by PLS, accompanied by a noticeable pELS fraction and only minor dELS representation. Other cCRE classes, including CTCF-associated categories, were negligible.

Genic feature distributions were concordant with the cCRE-based assignments (Figure 4). H3K4me3 peaks were enriched at promoters, with smaller contributions from exons and introns, consistent with promoter-proximal chromatin. H3K4me1 peaks were comparatively depleted at promoters and shifted toward non-promoter features, with introns constituting the largest category and an additional other portion apparent; 5' and 3' UTRs were minor for both marks.

Together, these orthogonal annotations support the expected functional specializations of the marks: H3K4me3 exhibits a promoter-centric profile, whereas H3K4me1 preferentially

labels enhancer-associated regions, many of which reside within intronic and other non-promoter genomic contexts.

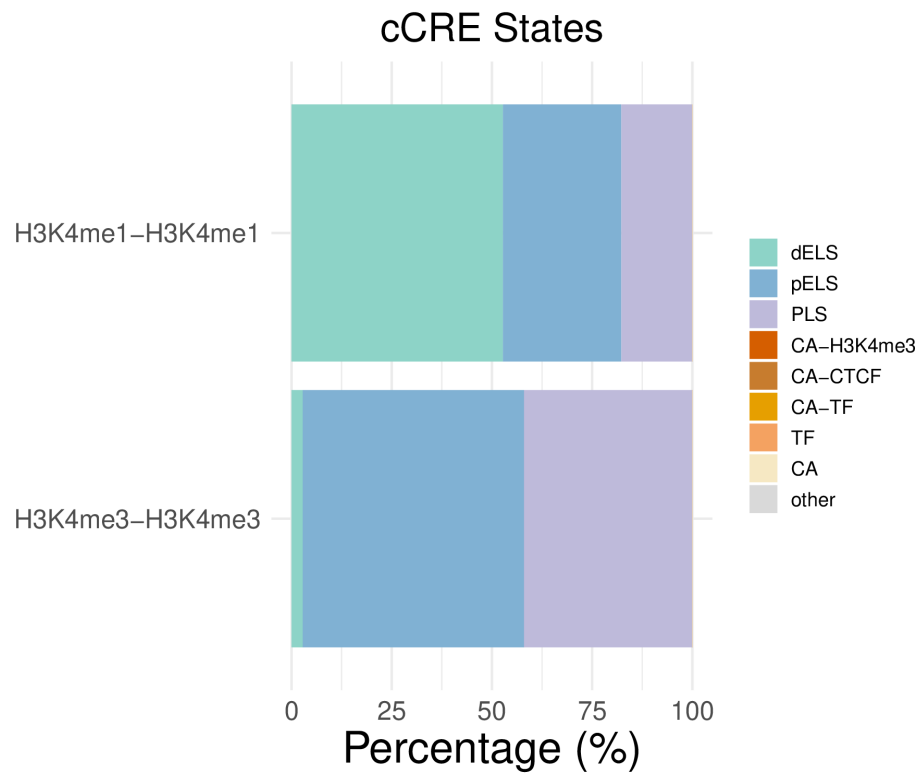


Figure 3. cCRE composition contrasts for H3K4me1 versus H3K4me3 peaks. *Called peaks were overlapped with cCRE annotations to assign classes. H3K4me1 peaks map mainly to enhancer-like (dELS, pELS) elements, whereas H3K4me3 peaks are largely promoter-like (PLS) with notable pELS and minimal dELS representation.*

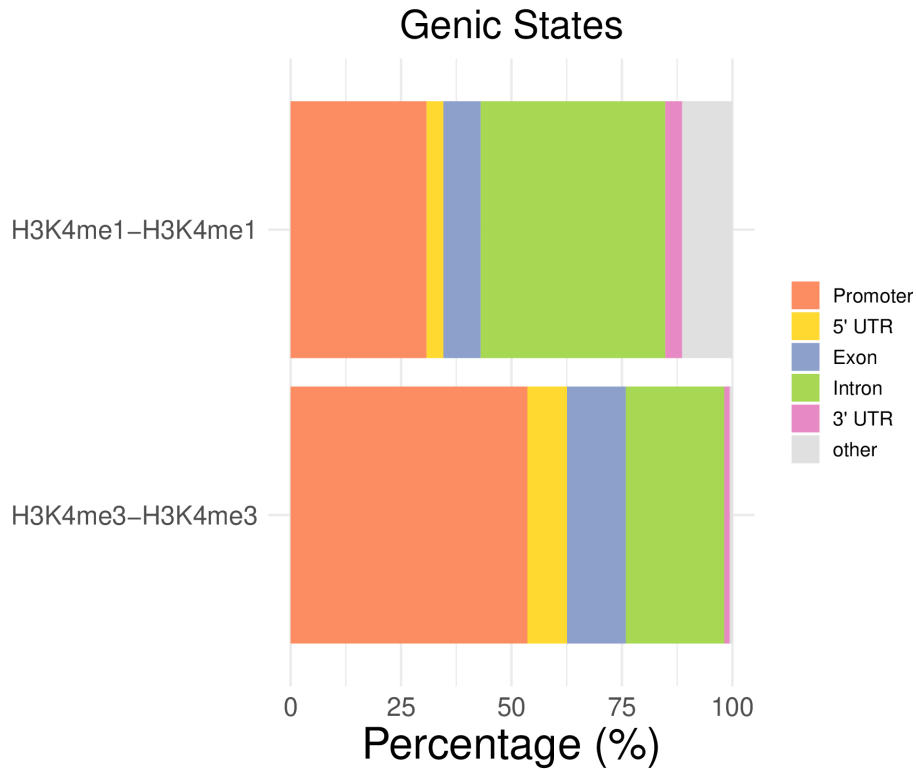


Figure 4. Genic distribution contrasts H3K4me1 and H3K4me3 peaks. Peaks were annotated to promoters, UTRs, exons, introns, and other regions. H3K4me3 maps mainly to promoters, while H3K4me1 shifts toward intronic and other non-promoter annotations; exonic and UTR fractions remain small for both marks.

Cross-mark signal relationships over the union peak set

We quantified cross-mark concordance by comparing read densities for H3K4me1 and H3K4me3 across the union peak set ($n=1,546$), using CPM values that were $\log_1 p$ -transformed. The scatter plot (Figure 5) reveals a weak but significant inverse relationship between the two marks (Pearson $r = -0.1686$, $p \sim 2.56e-11$), with points partitioning into mark-biased clusters: one enriched for H3K4me3 with H3K4me1 at moderate levels, and another enriched for H3K4me1 with comparatively low H3K4me3. Instances of simultaneous high signal for both marks were uncommon, and very low H3K4me1 values were rare. The 2x2 correlation heatmap (Figure 6) recapitulates this global trend, showing unity on the diagonal and an overall H3K4me1-H3K4me3 correlation of approximately -0.17.

These patterns are consistent with the distinct promoter/enhancer biases in this dataset: H3K4me3-dominant peaks correspond to promoter-like regions, whereas H3K4me1-dominant peaks reflect enhancer-like regions. The modest anticorrelation arises from reciprocal enrichment across these functional classes within the merged peak set, while its

weak magnitude indicates partial overlap and gradation in mark occupancy across loci rather than strict mutual exclusivity.

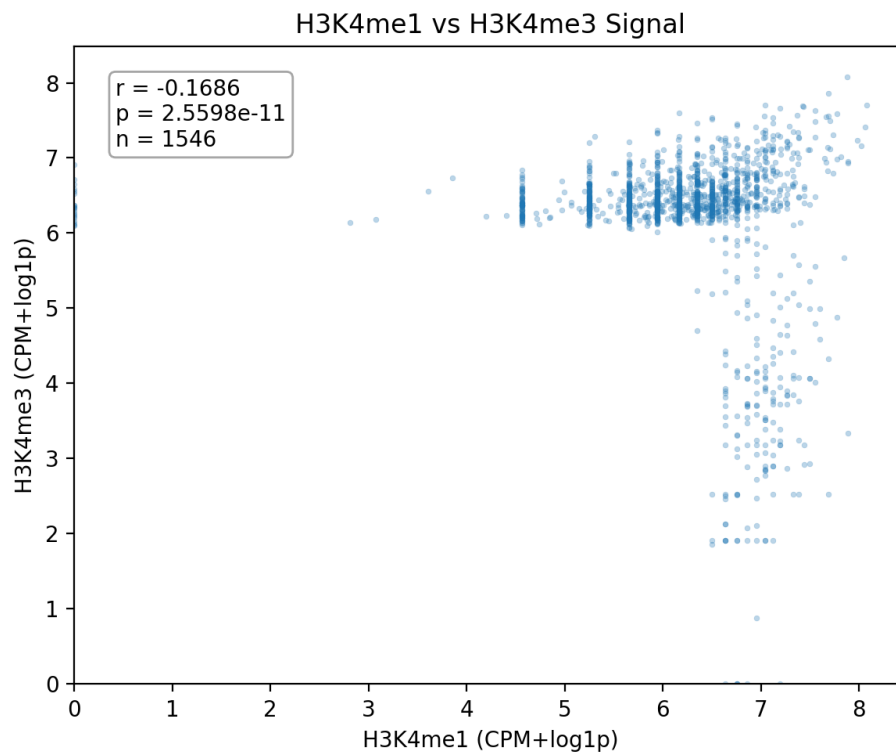


Figure 5. H3K4me1 and H3K4me3 signals show weak anticorrelation. Scatter of normalized read densities across 1,546 union peaks compares H3K4me1 (x) with H3K4me3 (y). Points favor mark-skewed regions over co-enrichment, yielding a weak negative correlation ($r = -0.1686$, $p = 2.56e-11$).

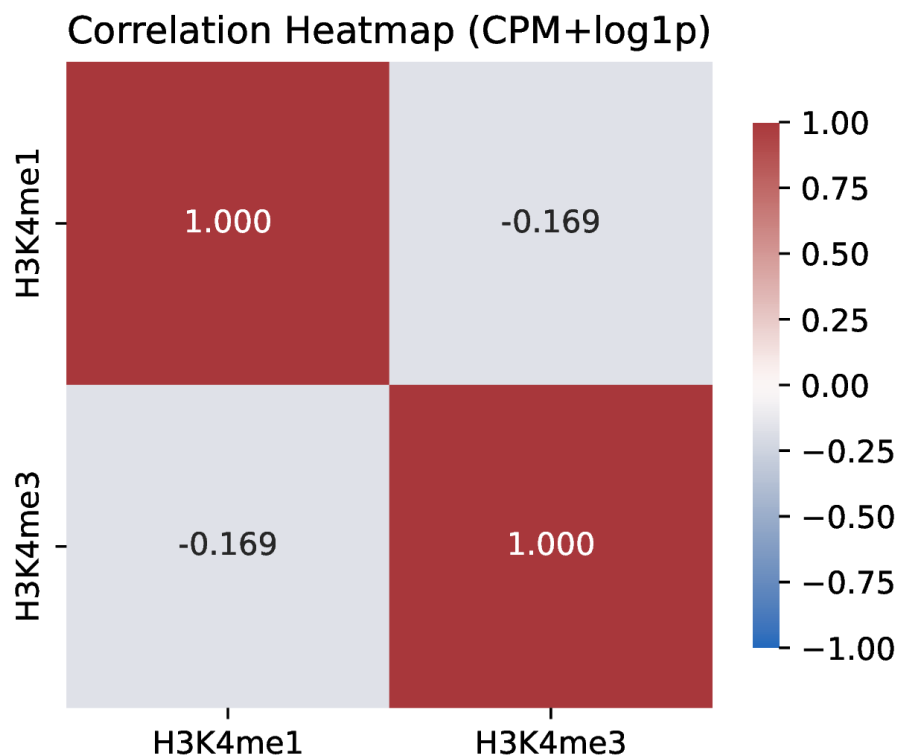


Figure 6. Pairwise correlations between H3K4me1 and H3K4me3 over peaks. *Pearson correlations were computed from CPM+log1p signals across the union peak set. Self-correlations are 1.000, and the cross-mark correlation is -0.169, indicating a slight negative association.*

References

1. Wen, K., Hu, T., Wang, Q., & Sun, X. (2026). Nucleophosmin 1 proteins as potential therapeutic targets in non-communicable chronic inflammatory diseases: a review of pathophysiological mechanisms. *Frontiers in Cell and Developmental Biology*, 14. <https://doi.org/10.3389/fcell.2026.1840902>
2. Camacho, L. Q., & Fallahi-Sichani, M. (2026). Systematic multivariate analysis of chromatin complex dependencies reveals Set1C/COMPASS as a melanoma-enriched epigenetic vulnerability. *PLOS Computational Biology*, 22(8), e1014018. <https://doi.org/10.1371/journal.pcbi.1014018>
3. Yang, Q., Vafaei, S., Falahati, A., Khosh, A., Omran, M. M., Bai, T., Bariani, M. V., Ali, M., Boyer, T. G., & Al-Hendy, A. (2025). Decoding Bromodomain and Extra-Terminal Domain Protein-Mediated Epigenetic Mechanisms in Human Uterine Fibroids. *International Journal of Molecular Sciences*, 26(24), 12144. <https://doi.org/10.3390/ijms262412144>
4. Ueno, M., Kubo, S., Todoroki, Y., Nakayamada, S., & Tanaka, Y. (2026). Immune regulation and cell metabolism in B cell subsets in patients with systemic lupus erythematosus. *Frontiers in Immunology*, 17. <https://doi.org/10.3389/fimmu.2026.1755589>

5. Liu, F., Wang, Q., Xiong, J., Wang, M., Zhou, H., & Xiao, Y. (2025). Parental S-adenosylmethionine diet defines offspring immune response via histone H3K4me3 complex and Endoplasmic Reticulum UPR. *Cell Communication and Signaling*, 23(1). <https://doi.org/10.1186/s12964-025-02386-7>